

C02 - ASSESSMENT TOOL - 1

ANALYTICAL PROBLEM.

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Course: Computer Vision.

Course code: CSA0502

Grey Level Transformation:

Grey level Transformation is an image enhancement techniques used to modify the intensity (grey level) values of an images to improve its visual quality. It is a point operation where each pixel is processed independently.

$$S = T(r)$$

a) Explain the purpose of grey level transformation

Grey level transformation is an images enhancement technique used to modify pixel intensity values and improve image Quality.

Purpose:

- Improves brightness & contrast
- Enhances image quality
- makes hidden details visible.
- Used in medical imaging, satellite images, & computer vision

Negative Transformation formula:

$$S = 255 - r$$

Where,

r = Original pixel value

S = Transformed pixel value.

b) Apply the Negative Transformation

Given:

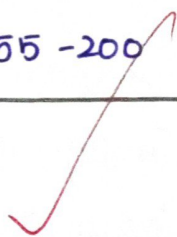
$$s = 255 - r$$

Original pixel values

$$r = 50, 100, 150, 200$$

Calculation:

Original Pixel (r)	Calculations	Transformed Pixel (s)
50	$255 - 50$	205
100	$255 - 100$	155
150	$255 - 150$	105
200	$255 - 200$	55



Original Pixel (r)	Transformed pixel (s)
50	205
100	155
150	105
200	55

Interpretations: -

- ★ Dark pixel becomes bright
- ★ Bright pixel become dark
- ★ The images appears like a photographic negative
- ★ Useful for highlighting details in dark region

Application:

- Medical imaging
- Satellite image processing
- Computer Vision

2. Histogram Equalization

Histogram Equalization is an image enhancement technique used to improve the contrast of a grayscale image by redistributing the intensity values over the entire available range. It spreads frequently occurring intensity values to make the images clearer.

a) Explain Histogram equalization :-

Histogram equalization improves image contrast by redistributing pixel intensity values.

Advantages :-

- ★ Improves image contrast.
- ★ Enhance hidden details.
- ★ Produces better intensity distribution

Intensity	frequency
0	4
1	6
2	10
3	6

Total pixel:

$$N = 4 + 6 + 10 + 6 \\ = 26.$$

Maximum Intensity -

$$L=4$$

b) Compute CDF

Intensity	Frequency	Cumulative Frequency	CDF
0	4	4	$4/26 = 0.154$
1	6	10	$10/26 = 0.385$
2	10	20	$20/26 = 0.769$
3	6	30	$26/26 = 1.000$

c) Equalized Intensities:-

formula

$$s = (L-1) \times \text{CDF}$$

Since

$$L=4, L-1=3.$$

Original	Equalized
0	0
1	1
2	2
3	3

Interpretation:-

- ★ Image contrast improves.
- ★ Hidden details becomes clearer.
- ★ Pixel values are more evenly distributed.

3 Gaussian Smoothing:-

a) Gaussian Smoothing is an image filtering technique used to reduce image noise & blur unnecessary details using a Gaussian function. It is given more important to center pixel than the surrounding pixel, resulting in smooth & natural-looking images.

Purpose:

- ★ Remove image noise
- ★ Smooths the image
- ★ Preserves important edges.
- ★ Images equally before edge detection

b) Apply Gaussian Kernels:-

Kernels

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Image:

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

$$(10 \times 1) + (20 \times 2) + (10 \times 1) + (20 \times 2) + (40 \times 4) + (20 \times 2) + (10 \times 1) + (20 \times 2) + (10 \times 1) \\ = 360.$$

= 360.

Interpretation:

Center pixel = 23

Noise is reduced.

Image becomes smoother while preserving edges.

4. Edge Detection using Sobel Operator.

- a) Edge detection is an image processing technique used to identify the boundaries of object by detecting sudden changes in pixel intensity. The Sobel operator calculates the gradient of images intensity to detect edges.

Purpose:

Detect object boundaries.

Image Segmentation

Pattern Recognition.

Features extraction.

- b) Apply horizontal Sobel Mask

mask =

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Images:

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Calculation :

Top row :

$$(-1 \times 10) + (-2 \times 10) + (-1 \times 10) = -40$$

Bottom row :

$$(1 \times 30) + (2 \times 30) + (1 \times 30) = 120$$

Gradient :

$$120 - 40 = 80$$

Final Answer $G_y = 80$.

Gradient value = 80

Indicates a strong edge

higher gradient means stronger intensity changes.

a) Contrast Stretching:

Contrast stretching is an image enhancement technique used to increase the contrast of an image by expanding the range of pixel intensity values. It improves image visibility by utilizing the complete grayscale range.

Purpose:

Improves image contrast

Enhances visibility

Makes details clear

Uses the full grayscale range.

Formula:

$$S = \frac{(r - r_{\min})}{(r_{\max} - r_{\min})} \times 255$$

b) Linear Contrast Stretching:-

Old range:

50-150

New Range:

0-255

Formula:

$$S = \frac{(r - r_{\min})}{(r_{\max} - r_{\min})} \times 255$$

Where

$$r_{\min} = 50, r_{\max} = 150.$$

Calculation:

for $\gamma = 50$

$$s = 0$$

for $\gamma = 100$

$$s = \frac{50}{100} \times 255 = 127.5 \approx 128$$

for $\gamma = 150$

$$s = 255$$

Original Intensity	Transformed intensity
50	0
100	128
150	255

Interpretation

Contrast is improved

Dark pixel becomes darker.

Bright pixel becomes brighter.

Images details become more visible.